

Music Performance & Streaming Analytics

1. Project Overview and Objective

The Music Performance and Streaming Analytics project focuses on analyzing music consumption data from multiple streaming platforms to understand song performance, audience engagement, and emerging trends. The project collects and evaluates key metrics such as streams, playlist placements, listener reach, Shazam counts, station plays, and platform-specific performance indicators from services such as Spotify, Apple Music, Amazon Music, Deezer, Pandora, and SoundCloud.

By applying data analysis and visualization techniques, the project provides insights into artist growth, track popularity, listener behaviour, and the effectiveness of music promotion strategies. The analytics dashboard helps stakeholders monitor performance trends, compare platforms, and make data-driven decisions for marketing, distribution, and audience development.

2. Data Sources

- **Domain:** Music Performance

3. Problem Statement

This project aims to analyze a music streaming dataset to identify the key factors influencing song popularity, compare performance across streaming platforms, evaluate artist and album success, and uncover trends based on release year and user engagement. The analysis will help answer business questions related to streaming performance, audience behavior, and the characteristics of successful songs through data cleaning, exploratory analysis, and interactive visualizations.

- Lack of real-time insights into music performance quality
- Artist wise spotify playlist count
- Number of spotify playlist count
- Number of time viewers clicked the like button
- Number of time tiktok videos have been watched
- Number of apple music playlist that contain a particular song

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Track	String	A track usually means one individual song

Album Name	String (Text)	The title of that collection
Artist	String (Text)	Name of the artist
Release Date	Date	Date of releasing
ISRC	String (Text)	Tracks music plays
All time Rank	Number	the position of a song, album, or artist compared with all others throughout history
Track score	Number	numerical value that measures how well a song (track) is performing.
Spotify Streams	String (Text)	each time a song is played for at least 30 seconds
Spotify Playlist Count	Number	number of Spotify playlists that include a particular song
Spotify Playlist Reach	Number	total potential audience of all the playlists that include a song
Spotify Popularity	Number	Spotify assigns to a track, album, or artist to indicate how popular it is at the current time
Youtube Views	Number	show how many times a song's official video, lyric video, or audio video has been watched
Youtube likes	Number	number of times viewers have clicked the Like button on a YouTube video
TikTok post	Number	video or piece of content uploaded to TikTok by a user, creator, or artist
TikTok likes	Number	number of times users have tapped the Like button on TikTok videos
TikTok views	Number	number of times TikTok videos have been watched
Youtube Playlist Reach	Number	potential number of viewers who can discover or watch a song

		through YouTube playlists that include it
Apple music playlist count	Number	number of Apple Music playlists that contain a particular song.
Air play spins	Number	AirPlay keeps showing a loading/spinning circle and does not finish connecting.
SiriusXM spins	Number	SiriusXM keeps showing a loading/spinning icon and does not start playing or finish connecting.
Deezer playlist count	Number	number of playlists associated with an account, artist, or search result.
Deezer playlist reach	Number	number of people who can potentially be reached through a placement on Deezer.
Amazon playlist count	Number	number of playlists on Amazon Music, there is no public page that shows a total “Amazon playlist count.”
Pandora streams	Number	number of times a song, album, or artist has been played on Pandora.
Pandora track stations	Number	number of Pandora stations generated from a particular track
Sound cloud streams	Number	number of times a track has been played on SoundCloud.
Shazam counts	Number	number of times a song has been identified using Shazam.

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
 - **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.
-

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created measures.

Track	Album Name	Artist	Release Date
Million Dollar Baby -Single	Million Dollar Baby -Single	Tommy Richman	26-04-2024
Not Like Us	Not Like Us	Kendrick Lamar	05-04-2024
I Like The Way You Kiss Me	I Like The Way You Kiss Me	Artemas	19-03-2024
Flowers	Flowers - Single	Miley Cyrus	01-12-2023
Houdini	Houdini	Eminem	31-05-2024
Lovin On Me	Lovin On Me	Jack Harlow	11-10-2023
Beautiful Things	Beautiful Things	Benson Boone	18-01-2024
Gata Only	Gata Only	Floymenor	02-02-2024
Danza Kuduro - Cover	Don Omer	Music Lab Jpn	06-09-2024
Band4Band (Feat. Lil Baby)	Band4Band (Feat. Lil Baby)	Central Cee	23-05-2024
I Had Some Help (Feat. Morgan Wallen)	I Had Some Help	Post Malone	05-10-2024
The Door	The Door	Teddy Swims	14-06-2024
Lunch	Hit Me Hard And Soft	Billie Eilish	17-05-2024
Like That	We Don'T Trust You	Future	22-03-2024
Bathroom Floor	Bathroom Floor	Kids With Buns	10-07-2022
Lala	Lala - Single	Myke Towers	22-03-2023
Fortnight (Feat. Post Malone)	The Tortured Poets Department	Taylor Swift	18-04-2024
Greedy	Greedy	Tate Mcrae	14-09-2023
Blue	Hit Me Hard And Soft	Billie Eilish	17-05-2024
As It Was	As It Was	Harry Styles	31-03-2022
Paint The Town Red	Paint The Town Red	Doja Cat	08-04-2023
Espresso	Espresso	Sabrina Carpenter	04-12-2024
Danza Kuduro - Cover	Club De Yokukiku Yougaku Medley	Music Lab Jpn	21-05-2024
Texas Hold 'Em	Texas Hold 'Em	Beyoncé	02-10-2024
Future	Future	Official Sound Studio	01-10-2024
Beat Automotivo Tan Tan Tan Viral	Beat Automotivo Tan Tan Tan Viral	Wz Best	24-05-2022

- Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

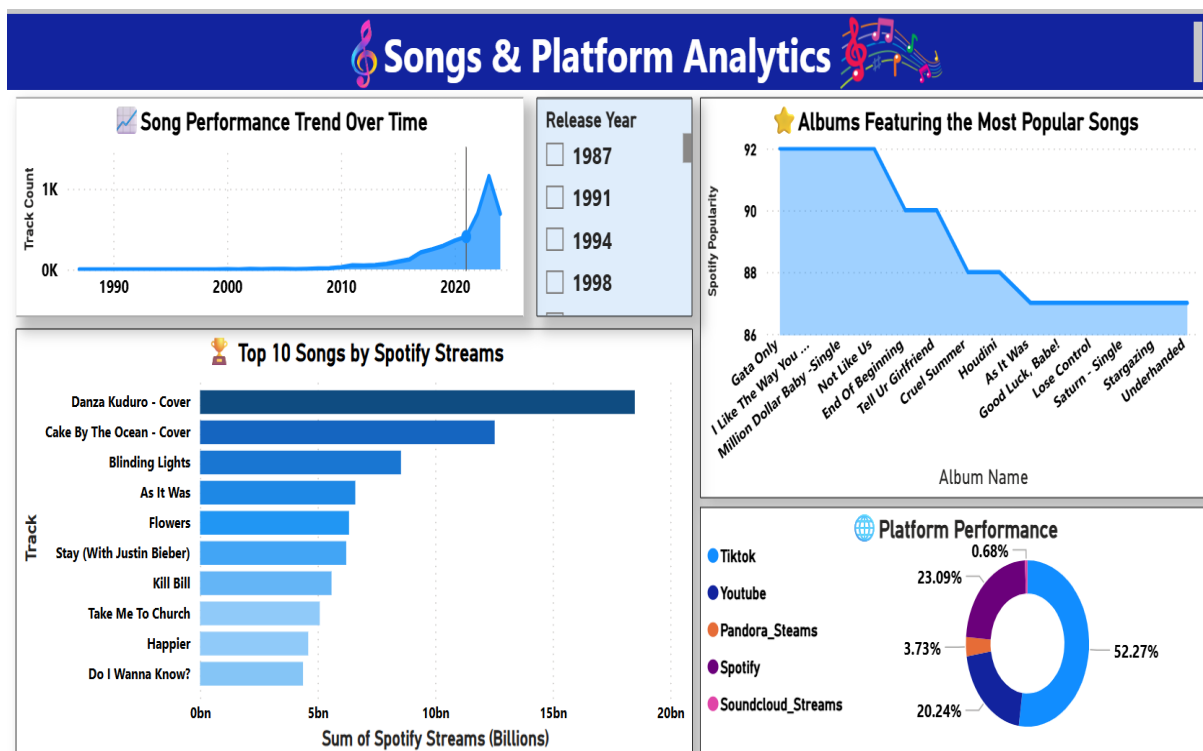
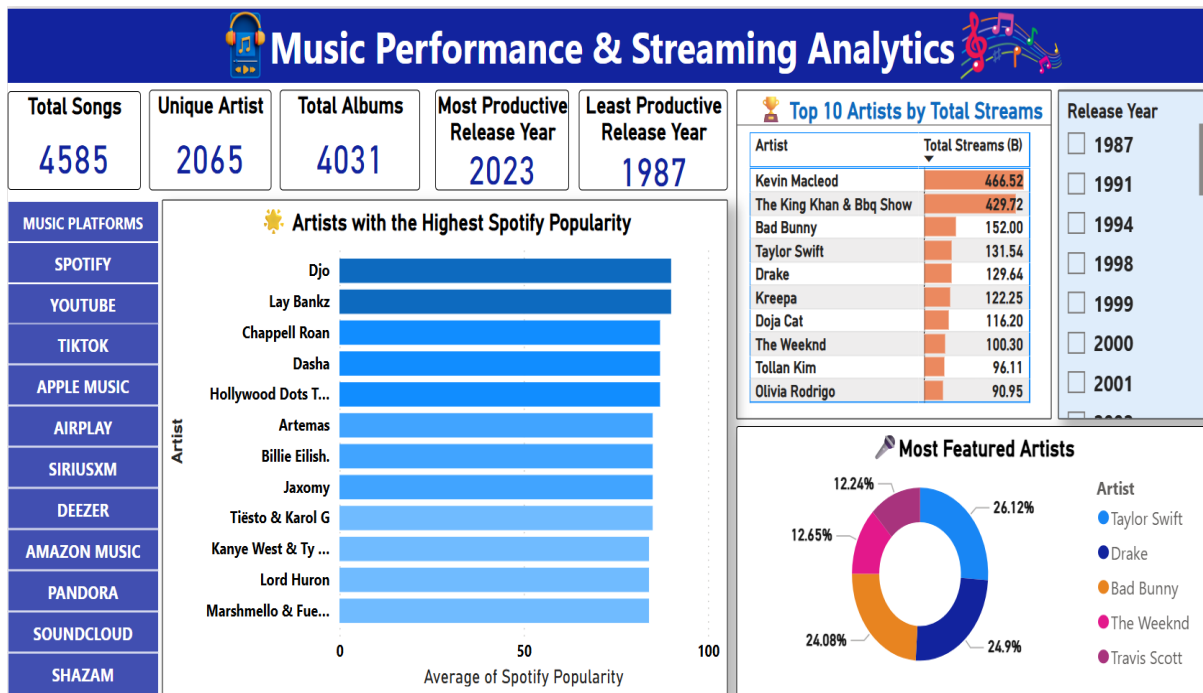
Top 10 Artist Performance		Most frequent artist in the dataset	
Artist	Total Views	Artist	Artist Count
Kevin Macleod	466.52B	Taylor Swift	64
The King Khan & Bbq Show	429.72B	Drake	61
Bad Bunny	152.00B	Bad Bunny	59
Taylor Swift	131.54B	The Weeknd	31
Drake	129.64B	Travis Scott	30
Kreepa	122.25B	Karol G	28
Doja Cat	116.20B	Ariana Grande	26
The Weeknd	100.30B	Billie Eilish	26
Tollan Kim	96.11B	Future	23
Olivia Rodrigo	90.95B	Post Malone	21
		Juice Wrld	21
		Peso Pluma	21
		Olivia Rodrigo	21
		Morgan Wallen	20
		Lil Baby	20
Highest Popularity			
Artist	Average of Spotify Popularity		
Lay Bankz	90		
Djo	90		
Chappell Roan	87		
Dasha	87		
Hollywood Dots The Oncelc	87		
Tiësto & Karol G	85		
Jaxomy	85		
Billie Eilish.	85		
Artemas	85		
Lord Huron	84		

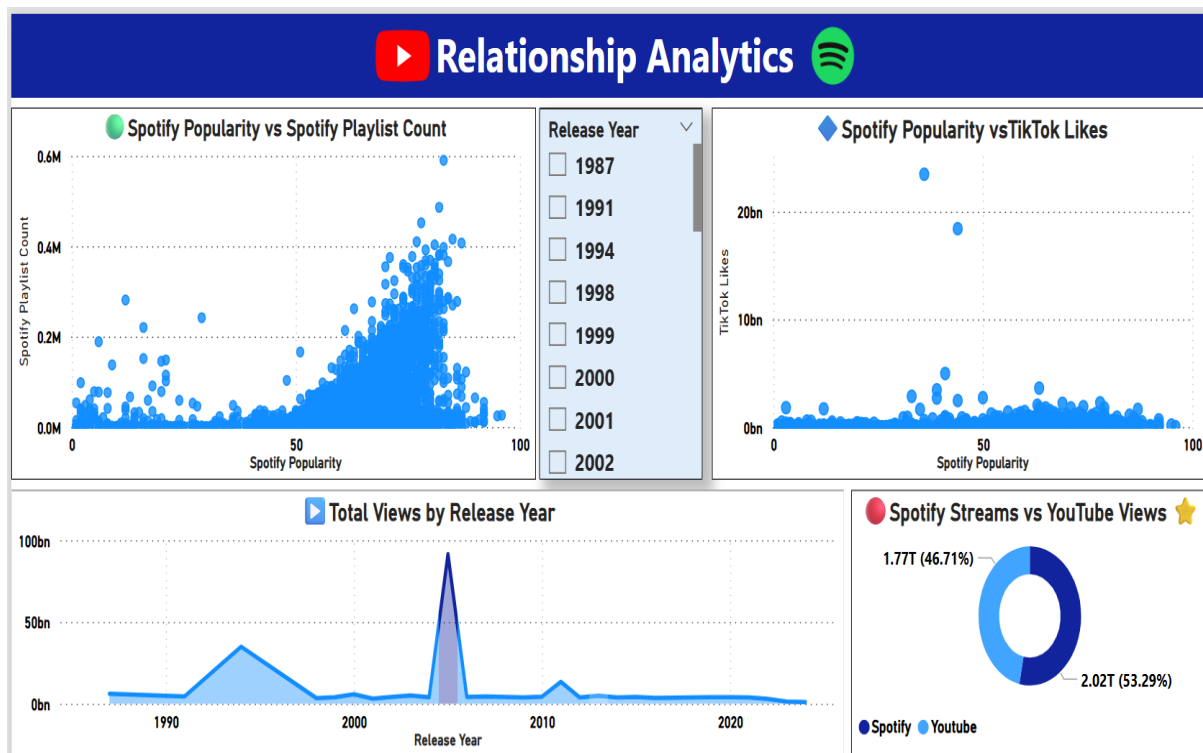
7. Data Modelling and DAX (Power BI)

- Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total songs, unique artists and total no. of albums

Total Songs	Unique Artist	Total Albums	Most Productive Release Year	Least Productive Release Year
4585	2065	4031	2023	1987

8. Analysis and Visualizations (Power BI)





Major Insights: Artist Performance

1. Audience Reach is Not Determined by the Number of Songs

Insight:

Kevin MacLeod ranks first in Total Views despite having only one track in the dataset, while Taylor Swift and Drake have many more songs but hold next place in the performance.

Decision:

Music companies should evaluate artists based on **audience engagement and content value**, not just the size of their music catalog.

2. Larger Catalogues Support Long-Term Growth

Insight:

Taylor Swift (64 songs), Drake (61 songs), and Bad Bunny (59 songs) have the largest catalogues and consistently rank among the top-performing artists.

Decision:

Record labels should encourage **consistent music releases** to build sustainable streaming revenue over time.

3. High-Quality Content Can Outperform High Quantity

Insight:

Artists with fewer songs, **such as Kevin MacLeod and The Weeknd**, generate exceptionally high audience reach, indicating that content quality and reuse are key success factors.

Decision:

Investment should prioritize **high-impact songs** with strong engagement potential rather than focusing only on increasing the number of releases.

4. Spotify Popularity Reflects Current Market Trends

Insight:

Artists like Lay Bankz and Djo have the highest Spotify popularity scores, showing strong current listener interest even though they are not among the highest in total views.

Decision:

Marketing teams should use Spotify Popularity to identify trending artists and launch timely promotional campaigns.

Song Performance – Major Insights

1. Songs Generate the Majority of Streams

Insight:

The Top 10 songs contribute a significantly larger share of total streams compared to the remaining songs in the dataset. This indicates that listener attention is concentrated on a few highly successful tracks.

Decision:

Streaming platforms should prioritize these high-performing songs in playlists and recommendations while also promoting emerging songs to diversify listener engagement.

2. songs have the highest popularity

Insight:

"Danza Kuduro - Cover" is the most popular track with a **Spotify Popularity Total of 824**, significantly outperforming all other songs. The top two cover songs (**Danza Kuduro - Cover** and **Cake By The Ocean - Cover**) dominate the popularity rankings, indicating that cover versions can achieve substantial audience engagement. The remaining tracks have popularity scores ranging from **213 to 277**, showing a noticeable gap between the top two songs and the rest.

Decision:

Marketing teams should use Spotify Popularity to identify trending songs early and increase promotional efforts while audience interest is growing.

3. Albums contain the most popular songs

Insight:

Four albums—**Not Like Us**, **Million Dollar Baby - Single**, **Gata Only**, and **I Like The Way You Kiss Me**—share the **highest average Spotify popularity score of 92**, making them the most popular albums in the dataset. Most of the top-performing albums have popularity scores between **87 and 92**, indicating consistently high audience engagement with only a small difference among the leading albums. The narrow popularity range suggests that several albums are performing competitively rather than one album dominating the rankings.

Decision:

Prioritize the top-rated albums (**92 popularity score**) for playlist placements, promotions, and marketing campaigns to maximize listener engagement.

Music companies should monitor social media trends and invest quickly in songs showing viral growth to maximize commercial opportunities.

Platform Analysis – Major Insights

1. Platform has the highest audience reach

Insight

TikTok has the highest audience reach with 4.58 billion views, more than double the reach of Spotify. Spotify ranks second with 2.02 billion streams, followed by YouTube with 1.77 billion views, indicating these platforms remain major drivers of music consumption. Pandora (327.37 million) and SoundCloud (59.38 million) contribute a comparatively smaller share of the total audience reach.

Decision

Prioritize TikTok for music promotions, viral campaigns, and influencer collaborations to maximize audience reach.

Maintain a strong presence on Spotify and YouTube to sustain streaming performance and listener engagement.

Use Pandora and SoundCloud as supporting platforms for niche audiences rather than primary marketing channels.

2. Platform contributes the most to song popularity

Insight

The dataset accumulated 290,879 Spotify popularity points and generated an impressive 2.02 billion Spotify streams, indicating strong overall audience engagement.

The high streaming volume suggests that Spotify is a major platform for music consumption across the analyzed tracks.

While popularity scores measure audience interest, the large number of streams reflects sustained listening behaviour.

Decision

Continue prioritizing Spotify as a key distribution and promotional platform due to its significant contribution to total streams.

3. Spotify, YouTube and TikTok performance

Insight:

Spotify records the highest streaming activity among all platforms, making it the dominant platform for music consumption in the dataset.

YouTube generates a large number of views and likes, showing that audiences prefer consuming music through video content as well as audio streaming.

Songs with higher TikTok posts and views often experience increased popularity across other streaming platforms, indicating TikTok's influence on music discovery.

Decision:

Record labels should adopt a **multi-platform marketing strategy** instead of relying on a single platform to maximize audience reach and reduce platform dependency.

Trend Analysis – Major Insights

1. Music Streaming Has Increased Over Time

Insight:

The analysis shows an overall upward trend in music streams over the years, indicating continuous growth in digital music consumption.

Decision:

Record labels should prioritize digital streaming platforms and allocate more resources to online music distribution, as consumer preference is shifting from traditional media to streaming services.

2. Recent Releases Gain Faster Audience Attention

Insight:

Recently released songs achieve high streams and popularity within a shorter period due to digital marketing, playlist recommendations, and social media exposure.

Decision:

Marketing campaigns should focus on the **first few weeks after a song's release**, as early promotion significantly influences long-term performance.

3. Streaming Performance Fluctuates Across Time

Insight:

The trend shows noticeable peaks and declines rather than continuous growth, indicating that audience engagement changes based on new releases, seasonal demand, and viral trends.

Decision:

Record labels should schedule major releases during periods of higher listener activity and maintain regular promotional campaigns during low-engagement periods.

Relationship Analysis – Major Insights

1. Spotify popularity and playlist presence

There is a clear concentration of songs with Spotify popularity around 60–80. Songs with higher popularity generally appear in more playlists, suggesting that playlist inclusion is strongly associated with visibility and popularity.

A few highly playlisted songs are clear outliers and may represent major hits.

2. Spotify popularity vs. TikTok likes

The relationship is not strongly linear. Some songs with relatively moderate Spotify popularity have extremely high TikTok engagement.

This indicates that TikTok virality does not necessarily depend on Spotify popularity.

TikTok can therefore be an important channel for discovering potential breakout songs.

3. Spotify streams vs. YouTube views

YouTube accounts for approximately 53.29% (2.02T) of the combined views/streams shown.

Spotify accounts for approximately 46.71% (1.77T).

The relatively close split indicates that both platforms are important for music consumption and reach.

Decision:

Prioritize songs based on **combined Spotify, TikTok, YouTube, and playlist performance**, rather than relying on a single platform metric.

10. Conclusions

Descriptive Analysis (What happened?)

The dataset contains 4,585 songs, 2,065 artists, and 4,031 albums, with a few artists and songs contributing the majority of streams. The dataset spans releases from **1987** to **2024**, enabling long-term trend analysis.

The descriptive analysis provides a comprehensive overview of the music streaming dataset by summarizing key performance indicators, identifying top-performing artists, songs, and albums, comparing audience reach across platforms, and highlighting release trends over time.

These findings establish a strong foundation for deeper diagnostic, predictive, and prescriptive analyses.

Diagnostic Analysis (Why did it happen?)

Spotify streams and YouTube views show a positive relationship.

Playlist inclusion helps improve song popularity.

Popular songs increase an album's overall performance.

Multi-platform presence increases audience reach and engagement

The number of song releases has increased steadily over the years, with 2023 emerging as the most productive release year but in 2024, songs release is slightly decreased.

Predictive Analysis

Historical trends suggest that popular artists, playlist inclusion, and promotion across platforms like Spotify, YouTube, and TikTok can help songs achieve better streaming performance and reach a wider audience in the future.

Prescriptive Analysis (What Should Be Done?)

Prescriptive analysis recommends actions based on the insights obtained from the descriptive, diagnostic, and predictive analyses. It helps identify strategies that can improve song performance, increase audience reach, and support better marketing decisions.

- 1) Promote songs across **Spotify, YouTube, TikTok, and other streaming platforms** to maximize audience reach and engagement.
- 2) Collaborating with well-known artists can help increase song visibility, attract new listeners, and improve streaming performance.
- 3) Since **Spotify and TikTok** contribute the largest audience reach, marketing efforts should prioritize these platforms to maximize impact.
- 4) Promote New Releases
- 5) Use streaming data to optimize future music marketing strategies